



Pimpri Chinchwad Education Trust (PCET)
Pimpri Chinchwad College of Engineering



Department of Information Technology

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**CipherDocs: A Secure, AI-Powered Solution to Prevent Forgery of
Critical and Official Documents Using Polygon Blockchain-Based
Encryption and Verification**

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Introduction of the topic

- Rapid growth of digital documents requires secure verification systems.
- Centralized systems are prone to forgery, tampering, and lack transparency.
- CipherDocs uses Polygon blockchain for tamper-proof and decentralized verification.
- Cryptographic hashing and encryption ensure document security and integrity.
- An AI validation layer (Vision-Language Models) enhances verification and usability.

Motivation

- Increasing document forgery and AI-generated fake certificates highlight security challenges.
- Centralized systems lack transparency and depend on manual verification.
- Need for decentralized trust to ensure long-term document integrity.
- Requirement for instant, secure, and privacy-preserving verification.
- Need for a low-cost, fast, and scalable solution using blockchain.

Literature Survey

Sr. No.	Paper Name	Core Technology	Specific Limitations Identified	CipherDocs Solution / Novelty
1	Certifier DApp – Decentralized and Secured Certification System Using Blockchain (Jha et al., 2023)	Ethereum DApp	High gas fees, only hash-based validation, no AI	Uses Polygon for low-cost transactions and AI for enhanced validation
2	DOC-BLOCK: A Blockchain-Based Authentication System for Digital Documents	Ethereum + IPFS	Binary validation, no explanation, privacy concerns	Adds explainable AI and encryption for secure and meaningful verification
3	A Comparative Study of Ethereum and Polygon for Implementing NFT-Based Certification Systems (Aung et al., 2024)	Eth vs Polygon	High cost and scalability issues in Ethereum	Uses Polygon for faster transactions and improved scalability
4	Docschain: Blockchain-Based IoT Solution for Verification of Degree Documents	IoT	OCR errors, sensitive to image quality and format limitations	Uses AI-based validation to improve accuracy and robustness

Literature Survey

Sr. No.	Paper Name	Core Technology	Main Findings	CipherDocs Solution / Novelty
5	A Systematic Literature Review on Blockchain-Based Systems for Academic Certificate Verification (Rustemi et al., 2023)	Blockchain systems	Focus on security, lacks usability and automation	Provides user-friendly system with AI-assisted verification
6	Decentralized Certificate Issuance and Verification System Using Ethereum Blockchain Technology (Sree, 2025)	Ethereum + Smart Contracts	High gas cost and performance limitations	Improves efficiency using Polygon and optimized workflow
7	Decentralized Document Storage and Retrieval Using Blockchain, IPFS, and QR Code Technology (Tengse et al., 2025)	Blockchain + IPFS + QR	Static QR codes, basic verification	Uses dynamic QR codes for real-time verification
8	Document Forgery Detection: A Comprehensive Review (Sukhija et al., 2025)	ML-based detection	No blockchain integration, lacks trust layer	Combines AI with blockchain for secure and intelligent verification

System Architecture

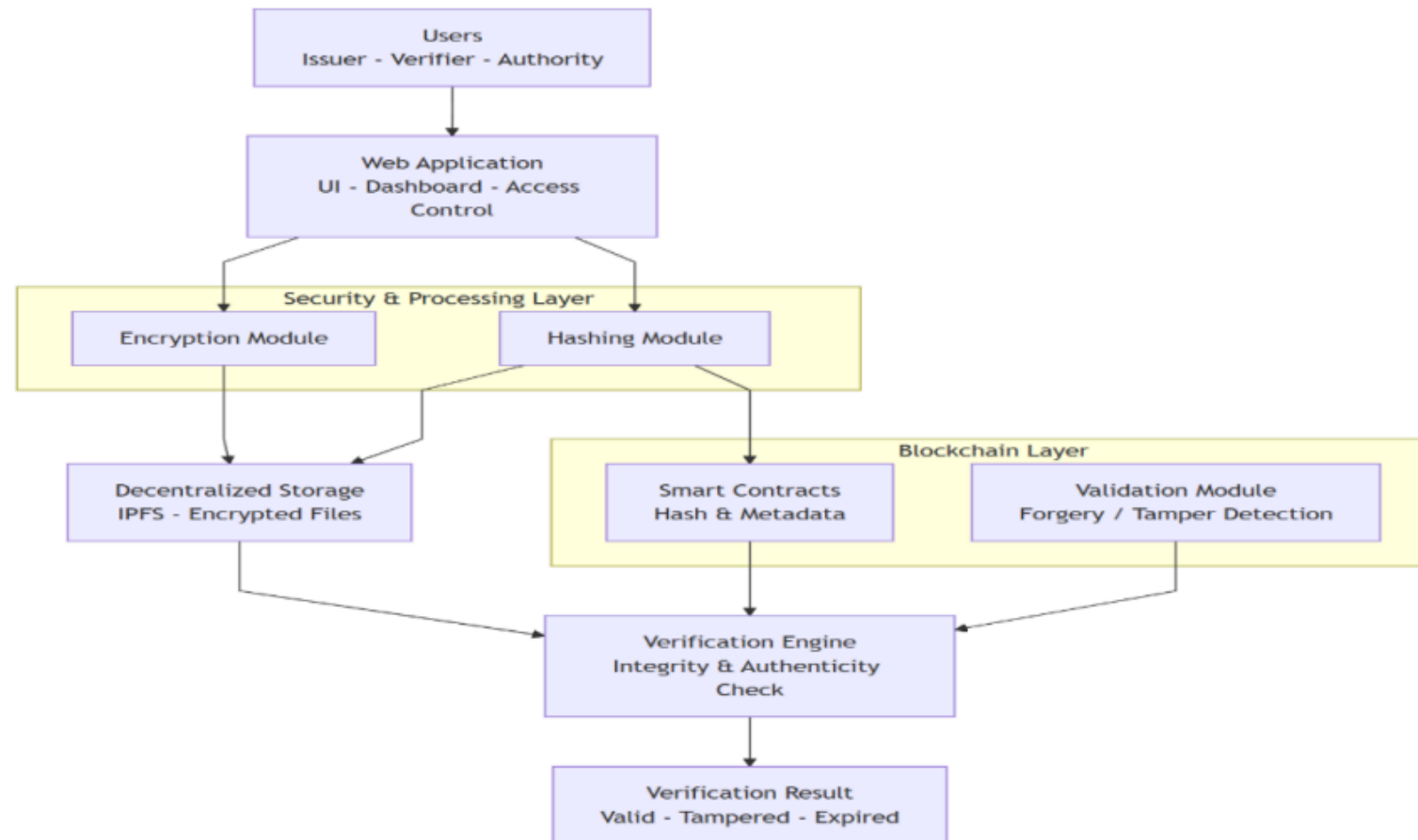


Figure. 1. Architecture Diagram

Flow Diagram

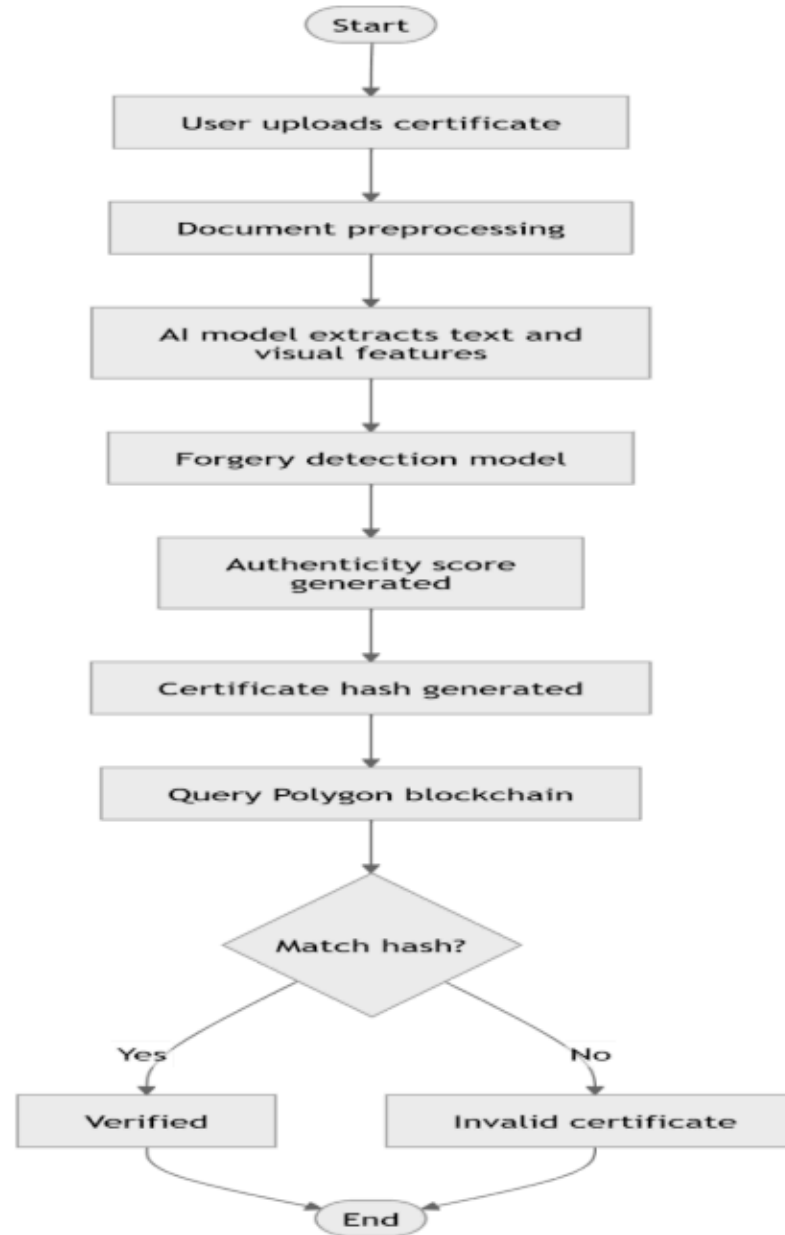


Figure. 2. Flow Diagram

Advantages & Applications

- Ensures tamper-proof and transparent verification using blockchain layer.
- Provides low-cost, fast, and scalable transactions using Polygon network.
- Eliminates manual verification through automated smart contract validation.
- Enhances verification using AI validation layer (Vision-Language Models).
- Applicable in education, government, land records, and enterprise systems.

Results

- Transaction cost is very low ($\sim \$0.001$) using Polygon, ensuring cost efficiency.
- Verification time is low ($\sim 150\text{--}250$ ms), enabling fast and reliable validation.
- System demonstrates high scalability for handling large-scale document verification.
- Hash-based validation ensures high accuracy in detecting any document tampering.
- AI validation layer (Vision-Language Models) provides high improvement in verification accuracy and clarity.

✓ Certificate is valid and authentic. VERIFIED

ISSUER WALLET ADDRESS
0xcae0a1144fe58397795e422786f076fa9c5a721c

USER WALLET ADDRESS
0xf0771b82c234222fb00b5a86a5800348f114e8BD

CERTIFICATE ISSUED AT
3/13/2026, 3:08:19 PM

BLOCKCHAIN TX
0x85b7f89abe329bb6518691de985d22caf44649ca570c0a85c9878506603cb97a

VERIFIED FILE
syOriginalResult.pdf

160.3 KB

! Document has been tampered. TAMPERED

Uploaded document does not match the originally issued certificate.

VERIFIED FILE
sy-fake.pdf

210.4 KB

MetaMask

Issuer

Transaction request

NetworkA Amoy

Request from ? ciphdocs.vercel.app

Interacting with ? ? 0x9d368...Aa467

Network fee ? < \$0.01 P POL

SpeedSite suggested ~2 sec

AI-Powered Analysis VERIFIED

Summary
The uploaded certificate has been verified against the original certificate stored in ciphdocs, and the result indicates that the certificate is valid.

Verification Status
The status returned by ciphdocs is "valid", indicating that the uploaded certificate matches the original certificate stored in the system.

Document Analysis
A thorough analysis of the uploaded certificate text and the original certificate text stored in ciphdocs has been conducted. The comparison reveals that the two documents are identical, with no discrepancies or modifications detected.

Conclusion
Based on the analysis, it can be concluded that the uploaded certificate is authentic and has not been tampered with. The certificate matches the original record stored in ciphdocs, and its validity has been confirmed. The verification result of "valid" indicates that the certificate can be trusted as a genuine document issued by the respective institution.

AI-Powered Analysis TAMPERED

Summary
The uploaded certificate has been marked as tampered by the ciphdocs verification system. This indicates that the certificate has undergone modifications after it was issued.

Verification Status
The status returned by ciphdocs is "tampered", suggesting that the uploaded certificate does not match the original certificate stored in the system.

Document Analysis
A comparison of the uploaded certificate text with the original certificate text stored in ciphdocs reveals several discrepancies. While the overall structure and content of the two certificates are similar, there are notable differences in certain fields.

Detected Differences
The following differences were detected between the uploaded and original certificates:

- The "ABC ID" field has been modified from "948128680857" to "949128680857".
- The "Year" field in the course "BIT440I" has been changed from "3" to "2".
- The "Course Name" for course code "BIT340I" has been altered from "DISCRETE MATHEMATICS" to no matching course, but "BIT340I" is associated with "LOGIC DESIGN AND COMPUTER ORGANIZATION" in the original, and "DISCRETE MATHEMATICS" is found in the uploaded text but associated with a different course code in the original.
- The "Cumulative CGPA" has been changed from "7.6" to "8.0".

Conclusion
Based on the analysis, it is clear that the uploaded certificate has been tampered with. The modifications detected suggest an attempt to enhance the academic performance or alter personal details. Therefore, the trust status of this document is compromised, and it should not be considered authentic. The ciphdocs verification system correctly identified the certificate as tampered, ensuring the integrity of the verification process.

Conclusion

- Establishes a trust-driven system to eliminate document forgery through decentralized verification.
- Enables automated and reliable validation, reducing manual dependency.
- Achieves cost-efficient and scalable performance using Polygon.
- Enhances verification with AI-based validation for better accuracy.
- Provides a secure and practical solution for modern document management.

References

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Thank You